THESIS PROJECT
TITLE PROPOSAL
(CS0025 – Software Engineering 1)

Group Name: **Statisfied**	Program: **Bachelor of Science in Computer Science with specialization in Data Science**	
Member's Name: 1. Dawn Andrei Pamesa (Project Manager) 2. Rogelio Porras (Member 1) 3. John Ashley Alba (Member 2) 4. Ralph Blanco (Member 3)	Term: [✓] 1st [ ] 2nd [ ] 3rd	Academic Year: 2026 - 2027
Proposed Project Title 2: **Weekly Construction Material Demand Forecasting for a Construction Firm Using a Cost-Sensitive Gradient Boosting Loss**		

1.0 Area of Investigation:

Construction firms depend on a steady supply of cement, sand, gravel and steel. When the weekly order is too small, work can be held up while the firm waits for a delivery. When it is too large, money is tied up in stock that sits in the warehouse. Inaccurate demand predictions based on traditional practice often lead to material waste, cost overruns and delays (Magableh et al., 2026).

The firm in this study is a Valenzuela construction firm with 34 years in business. It builds hospital buildings, office extensions, roads, slope protection and water works for private and government clients, and its main client's projects start at 50 million pesos. In its questionnaire answers the firm named materials needed as the hardest thing to predict. It buys mostly cement, sand, gravel and steel from 5 suppliers, deliveries take about 2 days, and the weekly order is approved once a week. It keeps its stock in an Excel inventory and checks it with weekly physical counts.

The firm keeps about 5 years of material records in Excel, and it confirmed to the group that the file holds around 10,000 rows. A row is not the same as a modeling sample, so the group will report the raw rows, the items, the observed weeks and the usable modeling observations separately once the

file is received.

The firm's three problems, in its own words, and the one chosen:

Problem	What the firm said	Status
1. Late collection and cash flow	Collection from the main client takes long; late payments and cash flow cause the most delay and loss.	Not chosen for this study. The firm's billing records cover only about 2 years and about 500 billings, far below the 10,000 data samples that the Data Science requirement asks for. Billing and payment dates may be used later as optional predictors.
2. Materials needed	Hardest to predict; the weekly order is approved once a week; main materials are cement, sand, gravel and steel; 5 suppliers; deliveries take about 2 days.	CHOSEN. The firm has about 5 years of material records, and the outcome (the weekly quantity issued to the site) can be checked against the firm's own records.
3. Cost and schedule changes	Final cost differs from the estimate; permits, rain, changed decisions and site obstructions.	Not chosen. Project change records may be used as optional extra predictors.

Payroll and manpower were also named, but they involve personal data and are outside the scope.

The analytics type is predictive (forecasting the weekly quantity of each material issued to the site), with descriptive summaries of past demand and price changes. The firm's preferred type will be confirmed in the client interview.

Algorithm/Technique Justification

Identified problem	Materials needed each week are hard to predict, so the weekly order depends on experience, and the firm risks shortages that can hold up work or extra stock that ties up its cash.
Computational task	Forecast the quantity of each material for next week (regression on a weekly time series for each item; the target is the quantity issued, as told by the firm), and measure shortage and excess exposure in pesos.
Proposed technique	To be confirmed by experiment. The leading candidate is gradient boosting (XGBoost) with a cost-sensitive loss that penalizes running short more than having extra, and weights items by unit price (Bauer et al., 2024; Laimighofer et al., 2022; Masnadi-Shirazi & Vasconcelos, 2011; Newey & Powell, 1987). No new algorithm is claimed.
Comparison	Last week's quantity, the average of the last four weeks, the same week last

	year, SARIMA, Random Forest, and XGBoost with the standard loss, on the same origins, target and test weeks, all described in 2.0.
Required data	About 5 years of the firm's material records in Excel, around 10,000 rows as told to the group, with the date, item, quantity and unit price. The adviser's rule of at least 5,000 rows from 2021 onward and the Data Science size of 10,000 data samples are checked against the file when it arrives, and the raw rows and the usable item-weeks are reported separately.
Expected output	A forecast of next week's quantity for each material, accuracy results, shortage and excess in pesos, and a plain comparison of the modified and unmodified models.
Evaluation	Rolling-origin testing with the first years for training, a validation year, and a final untouched test year (about 60% / 20% / 20% by date). MAE, RMSE and WAPE for the quantities of each material; shortage and excess exposure in pesos, always together; forecast quantity coverage; no leakage.

2.0 Background and Rationale of the Project/System:

Material demand in construction is hard to forecast because it depends on project schedules, weather and site conditions. Magableh et al. (2026) built machine learning models to forecast material demand, cost and arrival time from the data of three real projects in Jordan, and compared several regression models, including Random Forest and XGBoost. Fouad et al. (2025) studied AI-driven raw material demand forecasting for project management, and Egwim et al. (2021) applied artificial intelligence to predict construction project delay.

Prices matter too. The firm says material prices follow the market and high fuel costs. Liu et al. (2024) reviewed data-driven forecasting of construction material prices and grouped the models into causal models and time-series models. Ma et al. (2024) reviewed 34 studies on price forecasting in construction and found that vector error correction, artificial neural networks and ARIMA were the most used models. Gouda Mohamed et al. (2026) used vector autoregression on monthly prices of five construction materials, including steel and cement, together with macroeconomic indicators.

Outside construction, demand forecasting has many tested methods. Fildes et al. (2022) review retail forecasting, and Makridakis et al. (2022) report the results of the M5 forecasting competition. Tree-based methods often do well on tables of lagged values and calendar features. Haselbeck et al. (2022) found XGBoost the best in 14 of 15 comparisons for horticultural sales, Li et al. (2024) joined Random Forest and XGBoost to forecast short-cycle agricultural sales, Dankorpho (2024) applied XGBoost to retail sales, and Elorza et al. (2025) used a Prophet-XGBoost model for the demand of perishable products. Giannopoulos et al. (2025) review machine learning for intermittent

demand, which matters here because some items are ordered in some weeks and not in others.

A weekly material order has an uneven cost: a shortage can stop work, and extra stock costs money and space. Gradient boosting can be given a loss that treats the two mistakes differently. Bauer et al. (2024) propose boosting for regression quantiles with the pinball loss, which goes back to the regression quantiles of Koenker and Bassett (1978), Laimighofer et al. (2022) put an expectile loss into XGBoost to give extra weight to the lowest values when predicting low river flows, and Masnadi-Shirazi and Vasconcelos (2011) describe cost-sensitive boosting.

What the literature gap means. This gap is provisional until the search is documented. Based on the literature reviewed so far, studies forecast construction material demand, cost or price with standard models (Liu et al., 2024; Ma et al., 2024; Magableh et al., 2026). Limited evidence was found for a loss that penalizes shortage more than excess and weights items by their price, tested on a contractor's own weekly material records (searches done, not exhaustive). The asymmetric squared loss is closely related to expectile regression (Newey & Powell, 1987), so the group does not claim a first or a new algorithm; the contribution is using this loss for weekly material orders and evaluating it against the unmodified model. The thesis treats the benefit as a hypothesis and tests it directly.

Task and data. Ordered, delivered and issued quantities are different things, so the study uses one primary target. The firm told the group that its records include the quantity issued or used on site, so the primary target is the weekly quantity issued to the site, used as a proxy for the weekly material requirement, not as a verified measure of what was physically consumed. This is checked when the file is received. If it turns out that the records hold only orders or deliveries, the target is renamed after that quantity, for example the weekly quantity delivered, and the study does not claim to measure the quantity actually used. A purchase-order quantity is not called demand without that justification. The unit of analysis is one material in one week, and the target is that quantity in the next week. Unit prices from the firm's own records are used as predictors. The weekly order is approved on a fixed day, and that day is the decision time: a predictor or a past target is used only if it was already recorded by then.

How the loss is defined. This is the training loss, kept apart from the peso measures in the Metrics paragraph. Items are counted in different units (bags, cubic meters, kilograms and so on), so every item's quantities are first expressed in one fixed issue unit from the firm's records, and then divided by that item's mean weekly quantity in the training weeks only. The models E, F and G learn on these scaled quantities, and the forecasts are multiplied back by the same training mean for reporting and for the peso measures. For an item in a week, let the error be the forecast minus the actual quantity, on the scaled quantity. The loss is a weight times the squared error, and when the forecast is too low (negative error) the squared error is multiplied by a penalty k of at least 1. The weight is the item's unit price times its training mean weekly quantity (the peso value of one scaled unit of error), divided by the median of that value over all items, capped at an upper limit so that one very expensive item cannot dominate; the cap is itself tested. Setting k to 1 and every weight to 1 gives the ordinary squared loss, which is the unmodified control. The loss is implemented as a custom objective in the gradient boosting library (Friedman, 2001; for example XGBoost, Chen & Guestrin,

2016), with first derivative 2 x w x a x error and second derivative 2 x w x a, where a is k when the forecast is too low and 1 otherwise.

Setting k. The penalty k is not chosen by guess. If the firm gives its costs, k is set from its own cost ratio, that is the cost of one unit of shortage (delay or a rush delivery) divided by the cost of one unit of extra stock, using a balanced cost criterion fixed before testing. If the firm cannot give these costs, no financially best k is claimed. Instead k is evaluated over a fixed grid (1, 2, 3, 5 and 10) and the shortage and excess of every value are reported as a trade-off. If one setting is needed for the prototype, it is chosen on the validation weeks by a rule that is not financial and is fixed before testing: a coverage target is agreed with the firm (for example 90% of weeks), and the smallest k in the grid that reaches it is selected. If no k reaches the target, the result is reported as unmet and the k with the highest validation coverage is used. Excess exposure is always reported next to coverage, because high coverage can come from over-forecasting, and the selected k is never called financially optimal. One model is said to dominate another only when it is no worse on both shortage and excess and strictly better on at least one.

Models compared. A is last week's quantity. B is the average of the last four weeks. C is the same week last year (seasonal naive; Hyndman & Athanasopoulos, 2021). D is SARIMA. E is Random Forest (Breiman, 2001). F is XGBoost with the standard squared loss. G is XGBoost with the cost-sensitive loss, the leading candidate. G leads because of the studies above, not because it is confirmed. G is the proposed method of this study and A to F are the comparison models. G is evaluated and reported even if it does worse; the title names it because the study is built around it, and it can be made algorithm-neutral if the adviser prefers. All seven models are candidates, and the final technique and its settings are selected using a criterion fixed before testing, on validation results only: where the firm's cost coefficients are available, a balanced estimated cost; otherwise the validation WAPE is the primary accuracy criterion, the shortage and excess trade-off is reported separately, and any operational choice follows the nonfinancial rule above. The untouched final period only evaluates the selected technique, and negative or inconclusive findings are reported. To keep the comparison fair, every model uses the same historical information, the same forecast origins, the same target definition and the same test item-weeks. Random Forest, F and G use one identical feature set: lagged quantities, rolling averages, week of the year, month, holidays and unit prices, with project and weather information only if the records have them. The baselines A to C and SARIMA use only the material's own past quantities, which is how they are normally used. Models A to D are fitted for each material separately; E, F and G are each one model across all the items, with the item as a feature.

Loss ablation. To measure the price weights and the penalty separately, G is run in three controlled versions with the same data, features and evaluation: G1 uses the price weights with k = 1, G2 uses the penalty k with every weight equal to 1, and G uses both. F is the control with neither.

Training and testing. The data is split by date: about the first three years for training, the fourth year for validation (where k, the weight cap and the hyperparameters are fixed) and the fifth year as the final untouched test, that is about 60% / 20% / 20%. The training window expands as the

forecast origin moves forward (rolling origin). The model type, hyperparameters, loss settings and retraining schedule are frozen before the test year; if the models are retrained week by week during the test year, only outcomes and predictors known by that forecast origin are used. At each forecast origin a model is trained only on observations available by that date, and a training example is used only if the quantity of its target week was already recorded by the decision time of that origin, not merely dated before it. Lagged features and any scaling are calculated from the training weeks only. Random splitting is not used. Results are reported for each item.

Metrics. Three things are kept apart: the training loss (above), the peso exposure, and an estimated financial cost. Quantity errors: mean absolute error (MAE), root mean squared error (RMSE) and WAPE, the sum of absolute errors divided by the sum of actual quantities, which stays usable when some weeks are zero but is not defined for a material whose total actual quantity is zero. Because the items have different units, quantity errors are reported for each item, and for each group of items with the same unit, and are not added across units. The four main materials (cement, sand, gravel and steel) are also shown separately. Peso exposure: shortage exposure is the peso value of the quantity forecast too low and excess exposure is the peso value of the quantity forecast too high, using the unit price in the records; the two are always reported together. These are exposures, not realized costs. An estimated financial cost is calculated only if the firm gives its cost of shortage and of excess stock, and the coefficients are then stated and justified. Forecast quantity coverage: the share of weeks in which the forecast is at least the actual quantity. It is not called a service level, because no inventory fulfillment is modeled. Efficiency: training and prediction time. The metrics are to be agreed with the adviser and the firm.

Statistical check. The difference between G and F, and between G and the best simple baseline, is reported with a paired bootstrap confidence interval. The same test weeks are resampled for both models, in consecutive blocks of four calendar weeks taken across all items together, so that both models' results stay paired and common time effects are kept; item-weeks are not treated as independent, and the uncertainty is also reported for each item where there are enough weeks. The blocks are four weeks long (about one month, so that nearby weeks stay together; the length is checked against the autocorrelation of the errors). The best simple baseline is the one among A, B and C that is best on the validation weeks only. The difference and its uncertainty are reported, so that a small gain is not presented as a win.

Intermittent items. Some items are not ordered in some weeks. The share of zero weeks for each item is reported, and items with very many zero weeks are reported separately.

Data size. The firm told the group that the file has around 10,000 rows. That is the number of raw rows, not the number of modeling samples. A row may be a delivery line or a weekly stock record, and one material in one week may take several rows. The firm told the group that the file holds about 40 items over about 5 years (about 260 weeks). That is a theoretical maximum of about 10,400 item-weeks (40 x 260), not a verified count. About 96% of that grid would have to be present to reach 10,000, before any weeks are removed by cleaning, so the real number is likely to be lower. The study therefore reports the raw rows, the materials, the observed weeks, the item-weeks and the

usable training observations separately, and confirms with the adviser how the department counts a data sample. Rows are not duplicated or expanded to reach a number.

3.0 Statement of Objectives:

General objective. To develop and evaluate a gradient boosting model with a cost-sensitive loss that forecasts the weekly quantity of each main construction material for the firm, using the firm's own records, and to compare it with simpler forecasting methods.

Specific objectives. The specific objectives are as follows:

1. To build a cleaned weekly data set of the firm's material records (about 5 years, around 10,000 raw rows) with documented cleaning rules, and to report separately the raw rows, materials, observed weeks, item-weeks and usable modeling observations.
2. To set transparent baselines (last week, the four-week average and the same week last year) and to fit SARIMA, Random Forest and gradient boosting with its standard loss.
3. To implement the cost-sensitive loss, set its penalty from the firm's costs or report a fixed grid, and test whether it improves on the unmodified model, with the effect of the price weights (G1) and of the penalty (G2) measured separately from both together (G).
4. To evaluate all models on the untouched final year with accuracy metrics and with shortage and excess exposure in pesos, and to report where they fail.
5. To test, as a secondary objective and only if time allows, whether optional predictors (project schedule, rain, prices) add value, and to explain the forecasts in terms the firm's staff can use.

4.0 Scope and Limitations of the Study:

- One firm and the materials in its records, which hold about 40 items. Cement, sand, gravel and steel are the main ones. Items whose records are too incomplete to forecast are reported separately and left out of the model.
- About 5 years of the firm's own material records, around 10,000 raw rows as confirmed by the firm, with a forecast one week ahead because the order is approved once a week. About 40 items over about 260 weeks is a theoretical maximum of about 10,400 item-weeks; the number of usable item-weeks may be smaller and is reported as counted.
- Records must not contain personal data. Payroll and manpower records are out of scope. Names and prices of suppliers are hidden if the firm asks.
- The model predicts quantities only. It does not decide what to order or from which supplier; the firm's staff decide.
- Threats and how they are handled: an arbitrary penalty (set from the firm's costs, or reported

over a grid in a sensitivity table); leakage (only information known when the order is approved, time-based splits); items with many zero weeks (reported separately); a large new project that shifts demand (project information is used if the records have it); data quality, because records are kept in Excel, Viber, WhatsApp and on paper (only the Excel records are used, with documented cleaning rules); no improvement over the unmodified model (reported as a result).

- The study is not deployed live. The group hands over a working prototype.

5.0 Target Beneficiaries:

The firm's warehouse and procurement staff and management, through a clear forecast for the weekly order, which may help reduce shortages and excess stock (a possible benefit that this study does not measure).

The firm's project teams, through materials that arrive when the work needs them.

Other small and medium construction firms that could reuse the method.

6.0 SDG Category Number:



SDG Number: 8

SDG Title: Decent Work and Economic Growth

Justification: Target 8.3 (promote development-oriented policies that support productive activities, decent job creation, entrepreneurship, creativity and innovation, and encourage the formalization and growth of micro-, small- and medium-sized enterprises, including through access to financial services). The project helps a small construction contractor plan its weekly material orders from its own data, which may help it avoid shortages that hold up work and extra stock that ties up its money. The study does not measure that impact; it evaluates the quality of the forecasts.

7.0 References:

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Approved by:

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Panel Member 1 / Date

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Panel Member 2 / Date

Noted by:

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Thesis 1 Course Adviser / Date

1. AI Interaction Log (if applicable)

If your group used any Generative AI tool while completing the assessment, document how the tool was used. The log should demonstrate that AI supported the group's work rather than replacing the group's research, analysis, and decision-making.

Group Member	AI Tool Used	Prompt / Purpose	AI Output Used For	How the Output Was Verified or Modified
Dawn Andrei Pamesa	Claude (Claude Code)	Read the firm's questionnaire answers; drafted every section of this proposal, including the method (cost-sensitive loss, evaluation design); checked every cited paper on Crossref and read its abstract; formatted the document in the official layout.	Drafts of each section, reference checks, and formatting.	Every DOI was checked on Crossref and each statement about a study was matched to its abstract. The text was rewritten in the members' own words and checked against the sources.
	ChatGPT	Helped find sources.	A list of possible sources.	The sources were cross-checked.
	Custom GPT	Reviewed the draft twice as a thesis manuscript checker and listed problems to fix.	Two lists of problems in the target, data size, loss, evaluation, objectives and references.	We checked each point against the proposal and the sources, did our own research where needed, and applied only the valid ones.
Rogelio Porras	ChatGPT and Codex	Helped search for sources, and helped brainstorm and proofread his writing.	A list of possible sources, brainstorming ideas, and proofreading.	He read the sources himself before using them.

John Ashley Alba	ChatGPT	Helped gather sources and ideas, and helped check his work.	Suggestions of sources and ideas, and checking his work.	He read the sources himself and cross-checked them with other sites.
Ralph Blanco	None	None	None	None

2. AI Usage Declaration

We declare that any use of Generative AI in completing this Technical Assessment has been properly disclosed in the AI Interaction Log. We understand that AI is used only as an assistive tool and that our group remains responsible for verifying the accuracy, relevance, credibility, and originality of the submitted work.

We confirm that the final title proposal, objectives, scope, algorithm justification, and research outputs reflect our group's understanding and decisions. All literature, studies, and other external sources used have been properly cited and verified.

Group Name: Statisfied

Date: October 31, 2026

Group Member	**Signature**
Dawn Andrei Pamesa	
Rogelio Porras	
John Ashley Alba	
Ralph Blanco	

3. Team Contribution Matrix

The total contribution percentage must equal 100%.

Group Member	Assigned Task/Responsibility	Specific Contribution/Output	Contribution (%)
Dawn Andrei Pamesa	Project Manager; contacted V.M. Daet as the client and handled its questionnaire and data; planning with Ralph.	V.M. Daet client contact and questionnaire; planning and method design with Ralph; the proposal text, done with Claude.	30%
Rogelio Porras	Review of related literature and systems (RRL and RRS).	Sources and write-up for the RRL and RRS.	23%
John Ashley Alba	Review of related literature and systems (RRL and RRS).	Sources and write-up for the RRL and RRS.	23%
Ralph Blanco	Planning and brainstorming with the Project Manager.	Ideas and planning input for the proposal.	24%
		Total	100%